

Assignment 0

CSC-4780

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Task 1. Run ping between two instances.

```
ssh.cloud.google.com/v2/ssh/projects/nimble-sight-469821-c8/zones/us-central1-c/instances/instance-20250822-211815?authus...
SSH-in-browser
64 bytes from 10.128.0.3: icmp_seq=97 ttl=64 time=0.275 ms
64 bytes from 10.128.0.3: icmp_seq=98 ttl=64 time=0.329 ms
64 bytes from 10.128.0.3: icmp_seq=99 ttl=64 time=0.282 ms
64 bytes from 10.128.0.3: icmp_seq=100 ttl=64 time=0.230 ms
64 bytes from 10.128.0.3: icmp_seq=101 ttl=64 time=0.326 ms
64 bytes from 10.128.0.3: icmp_seq=102 ttl=64 time=0.288 ms
64 bytes from 10.128.0.3: icmp_seq=103 ttl=64 time=0.277 ms
64 bytes from 10.128.0.3: icmp_seq=104 ttl=64 time=0.273 ms
64 bytes from 10.128.0.3: icmp_seq=105 ttl=64 time=0.279 ms
64 bytes from 10.128.0.3: icmp_seq=106 ttl=64 time=0.352 ms
64 bytes from 10.128.0.3: icmp_seq=107 ttl=64 time=0.412 ms
64 bytes from 10.128.0.3: icmp_seq=108 ttl=64 time=0.441 ms
64 bytes from 10.128.0.3: icmp_seq=109 ttl=64 time=0.278 ms
64 bytes from 10.128.0.3: icmp_seq=110 ttl=64 time=0.267 ms
64 bytes from 10.128.0.3: icmp_seq=111 ttl=64 time=0.296 ms
64 bytes from 10.128.0.3: icmp_seq=112 ttl=64 time=0.304 ms
64 bytes from 10.128.0.3: icmp_seq=113 ttl=64 time=0.475 ms
64 bytes from 10.128.0.3: icmp_seq=114 ttl=64 time=0.304 ms
64 bytes from 10.128.0.3: icmp_seq=115 ttl=64 time=0.294 ms
64 bytes from 10.128.0.3: icmp_seq=116 ttl=64 time=0.358 ms
64 bytes from 10.128.0.3: icmp_seq=117 ttl=64 time=0.288 ms
64 bytes from 10.128.0.3: icmp_seq=118 ttl=64 time=0.326 ms
64 bytes from 10.128.0.3: icmp_seq=119 ttl=64 time=0.410 ms
64 bytes from 10.128.0.3: icmp_seq=120 ttl=64 time=0.284 ms
64 bytes from 10.128.0.3: icmp_seq=121 ttl=64 time=0.347 ms
64 bytes from 10.128.0.3: icmp_seq=122 ttl=64 time=0.249 ms
64 bytes from 10.128.0.3: icmp_seq=123 ttl=64 time=0.322 ms
64 bytes from 10.128.0.3: icmp_seq=124 ttl=64 time=0.368 ms
64 bytes from 10.128.0.3: icmp_seq=125 ttl=64 time=0.301 ms
64 bytes from 10.128.0.3: icmp_seq=126 ttl=64 time=0.337 ms
64 bytes from 10.128.0.3: icmp_seq=127 ttl=64 time=0.345 ms
64 bytes from 10.128.0.3: icmp_seq=128 ttl=64 time=0.435 ms
64 bytes from 10.128.0.3: icmp_seq=129 ttl=64 time=0.349 ms
64 bytes from 10.128.0.3: icmp_seq=130 ttl=64 time=0.331 ms
^C
--- 10.128.0.3 ping statistics ---
130 packets transmitted, 130 received, 0% packet loss, time 132060ms
rtt min/avg/max/mdev = 0.229/0.334/1.525/0.115 ms
chefmichaelserdar@instance-20250822-211815:~$
```

Field in the ping	Meaning
64 bytes from 1.0128.0.3	This is the size of the packet received from the IP address 10.128.0.3
icmp_seq=97	This is the sequences number of the request being sent. The number increases by 1 for each request.
ttl=64	ttl is the time-to-live and it is the max number of hops that the packet can take through the routers on the way to the destination
Time=0.275	This is the time that each packet takes to make a round trip between the two VMs

Summary	Meaning
130 packets transmitted	This is the total number of packets send while ping was running.
130 packets received	This is the total number of packets received by the second VM and as the numbers match it shows there was no loss.
0% packet loss	This shows that none of the packets were lost in transmission, which can also be seen in the above entries matching.
time 132060ms	This is the total time the ping was running in milliseconds.
rtt min/avg/max/mdev	This is the statistics of the session showing the minimum, average, max and mean deviation of the round trip times.

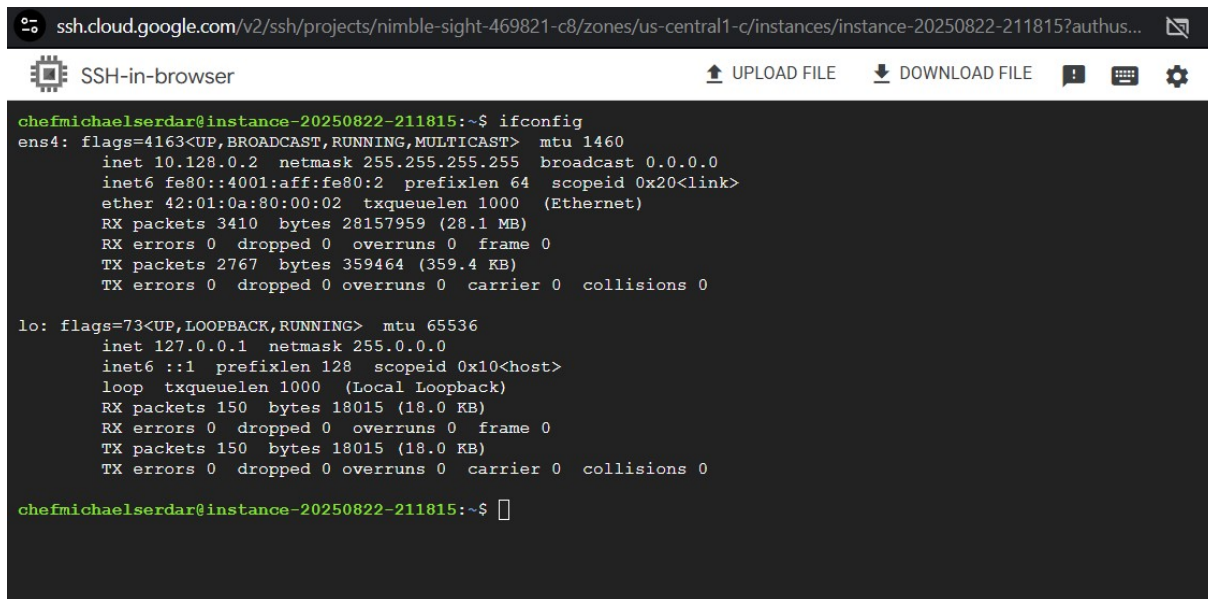
Task 2. Run traceroute and record the output.

```
ssh.cloud.google.com/v2/ssh/projects/nimble-sight-469821-c8/zones/us-central1-c/instances/instance-20250822-211815?authus...
SSH-in-browser
UPLOAD FILE
DOWNLOAD FILE

chefmichaelserdar@instance-20250822-211815:~$ traceroute tntech.edu
traceroute to tntech.edu (149.149.46.35), 30 hops max, 60 byte packets
 1 172.253.64.213 (172.253.64.213)  25.408 ms  24.659 ms  24.634 ms
 2 host-63-135-180-32.twlakes.net (63.135.180.32)  25.038 ms  24.294 ms  25.593 ms
 3 63.135.177.182 (63.135.177.182)  34.976 ms  33.235 ms  33.234 ms
 4 host-63-135-177-13.twlakes.net (63.135.177.13)  35.157 ms  34.665 ms  33.520 ms
 5 * * *
 6 149.149.2.2 (149.149.2.2)  33.827 ms  32.443 ms  32.035 ms
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
22 * * *
23 * * *
24 * * *
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
chefmichaelserdar@instance-20250822-211815:~$
```

Traceroute is used to follow the path your packets take through the different routers to reach the destination. The numbers listed are the IP addresses and the time it took to reach them. For the first entry 149.149.46.35 is the IP for Tntech.edu. 30 hops max means that traceroute will only try 30 routers max before stopping and 60-byte packets is the size of the packets being sent. Each hop after that is the IP address for the next router in the route and the time it took to reach it. This continues until the packets reach the destination. The asterisk in the output means that the packets were getting blocked by the firewall.

Task 3. Run ipconfig



The screenshot shows a terminal window titled "SSH-in-browser" with a URL bar indicating an SSH connection to a Google Cloud instance. The terminal displays the output of the `ifconfig` command for two network interfaces: `ens4` and `lo`.

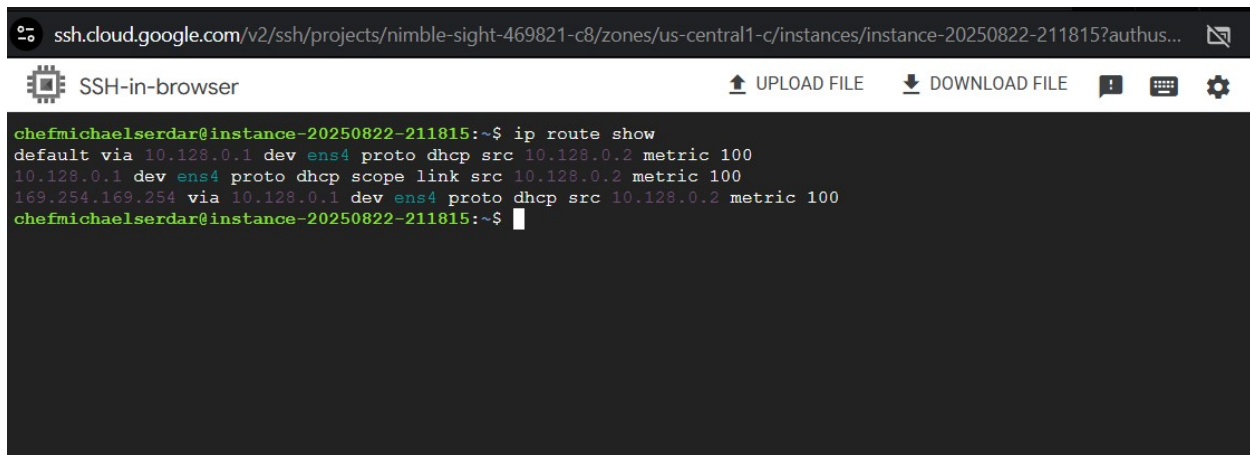
```
chefmichaelserdar@instance-20250822-211815:~$ ifconfig
ens4: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1460
    inet 10.128.0.2 netmask 255.255.255.255 broadcast 0.0.0.0
    inet6 fe80::4001:aff:fe80:2 prefixlen 64 scopeid 0x20<link>
    ether 42:01:0a:80:00:02 txqueuelen 1000 (Ethernet)
    RX packets 3410 bytes 28157959 (28.1 MB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 2767 bytes 359464 (359.4 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 150 bytes 18015 (18.0 KB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 150 bytes 18015 (18.0 KB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

chefmichaelserdar@instance-20250822-211815:~$
```

- **En24** is the modern Linux for the network interface, and it is used instead of eth0. En stands for ethernet and s4 means that it is attached to slot 4 on the PCI Bus
- **Inet 10.128.0.2** is the IPv4 address of the VM.
- **Netmask 255.255.255.255** is the subnet mask for the VM and this one means that there is only one address available.
- **Ether 42:01:0a:80:00:02** is the mac address for the actual hardware running this VM
- **Mtu 1460** is the largest packet size that can be transmitted

Task 4. Run ip route show



The screenshot shows a terminal window titled "SSH-in-browser" with a URL bar containing "ssh.cloud.google.com/v2/ssh/projects/nimble-sight-469821-c8/zones/us-central1-c/instances/instance-20250822-211815?authus...". The terminal output is as follows:

```
chefmichaelserdar@instance-20250822-211815:~$ ip route show
default via 10.128.0.1 dev ens4 proto dhcp src 10.128.0.2 metric 100
10.128.0.1 dev ens4 proto dhcp scope link src 10.128.0.2 metric 100
169.254.169.254 via 10.128.0.1 dev ens4 proto dhcp src 10.128.0.2 metric 100
chefmichaelserdar@instance-20250822-211815:~$
```

- **Default** means that this is being set as the default route that a packet will take if the IP doesn't match any other route.
- **Via 10.128.0.1** is the IP address of the router that sends messages outside the local network.
- **Dev ens4** is the name of the ethernet network interface used by the VM.
- **Proto dhcp** means that the route was set by the Dynamic Host Configuration Protocol or DHCP. This is used to automatically assign IP addresses in a local network.
- **Src 10.128.0.2** is the source address for this VM.
- **Metric 100** is the priority number for the route. The lower the number the better. This is used to decide which route is the best.